

Exam 1

Problems

Exercise 1 (1 point).

In regression analysis, the model in the form $y = \beta_0 + \beta_1 x + \varepsilon$ is called

- A. estimated simple linear regression equation
- B. simple linear regression model
- C. correlation equation

Solution: B

Exercise 2 (1 point).

A simple regression model has

- A. only one independent variable
- B. more than one dependent variable
- C. more than one independent variable
- D. at least two dependent variables

Solution: A

Exercise 3 (1 point).

In regression analysis, the variable that is being predicted is called the *response variable*. It is also called the

- A. dependent variable
- B. independent variable
- C. intervening variable
- D. usually x

Solution: A

Exercise 4 (1 point).

In simple linear regression (SLR), which of the following is **NOT** a required assumption about the error term ε ?

- A. The expected value of the error term is one.
- B. The variance of the error term is constant.
- C. The values of the error term are independent.
- D. The error term is normally distributed.

Solution: A

Exercise 5 (1 point).

A regression analysis between **sales** (measured in \$1000) and **price** (measured in dollars) resulted in the equation $\hat{y} = 60 - 8x$. This equation implies that an

- A. increase of \$1 in sales is associated with an increase of \$8 in price
- B. increase of \$1 in sales is associated with a decrease of \$52000 in price
- C. increase of \$1 in price is associated with an increase of \$52 in sales
- D. increase of \$1 in price is associated with a decrease of \$8000 in sales

Solution: D

Exercise 6 (1 point).

The confidence interval estimate for an **average value of y** (in a linear regression model) will be _____ compared to the prediction interval estimate for a particular value.

- A. the same
- B. wider
- C. narrower
- D. sometimes narrower, and sometimes wider

Solution: C

Exercise 7 (1 point).

In simple linear regression analysis, which of the following is **NOT true**?

- A. MSE equals SSE divided by the degrees of freedom for error.
- B. The F test and the t test have the same p-value, thus yield the same conclusion.
- C. The F test and the t test do not have the same p-value, thus do not yield the same conclusion.
- D. The relationship between x and y is represented by a straight line.

Solution: C

Exercise 8 (10 points).

At temperatures approaching absolute zero (-273 C), helium exhibits traits that seem to defy many laws of Newtonian physics. An experiment has been conducted with helium in solid form at various temperatures near absolute zero. The solid helium is placed in a dilution refrigerator along with a solid impure substance, and the fraction (in weight) of the impurity passing through the solid helium is recorded. (This phenomenon of solids passing directly through solids is known as quantum tunneling.) The data with 10 observations are given in the table.

Please fill in the blanks in the R output below.

Analysis of Variance Table

	Df	SS	MS	F	P
Source	1	0.83089	0.83089	46.728	0.00013289
Error	1.____	0.14225	2._____		
Total	9	0.97315			

Call:

```
lm(formula = PROPPASS ~ TEMP)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.30732	-0.02940	0.03045	0.05943	0.17014

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.490347	2.073772	-6.505	0.000187 ***
TEMP	-0.052829	0.007728	3._____	0.000133 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4._____ on 8 degrees of freedom
 Multiple R-squared: 0.8538, Adjusted R-squared: 0.8356
 F-statistic: 5.____ on 6.____ and 7.____ DF, p-value: 8.____

Solution:

1. 8
2. $0.14225/8 \approx 0.0178$
3. $-0.052829/0.007728 \approx -6.836$
4. $\sqrt{0.0178} \approx 0.133$
5. 46.728
6. 1
7. 8
8. 0.000133

Exercise 9 (3 points).

Jason believes that the sales of coffee (cups sold) at his coffee shop depend upon the temperature. He has taken a sample of 6 days. Some of the values were calculated based on the data collected.

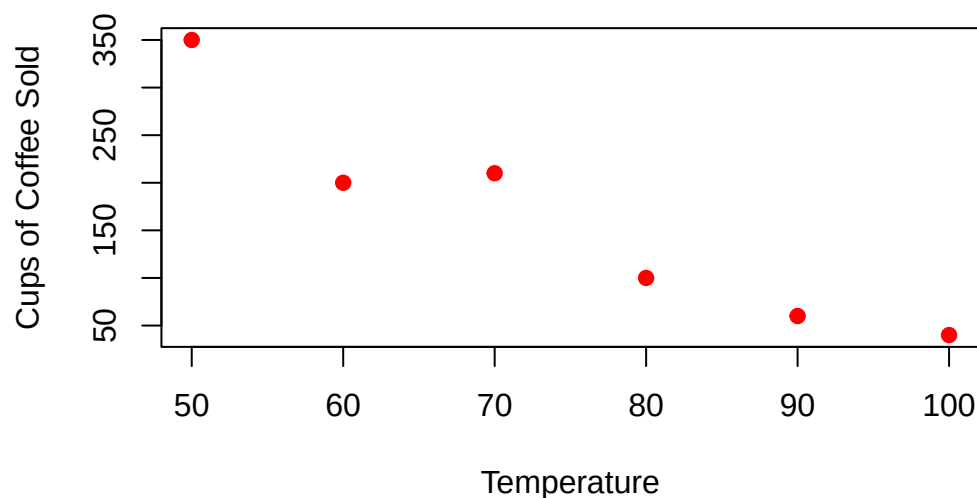
Use the following information for any calculations.

$$\bar{x} = 75, \quad \bar{y} = 160$$

$$SS_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = -10400, \quad SS_{xx} = \sum (x_i - \bar{x})^2 = 1750$$

$$SS_{yy} = \sum (y_i - \bar{y})^2 = 68200, \quad SSE = \sum (y_i - \hat{y}_i)^2 = 6394$$

Scatterplot of Cups of Coffee Sold vs Temperature



- a. What does the scatter diagram (above) indicate about the relationship between cups of coffee sold and temperature?
- b. What is the coefficient of determination r^2 ? (Display in percentage form, round to 1 decimal place.)
- c. Interpret the coefficient of determination r^2 .

Solution:

- a. They have an approximately negative linear relationship.
b.

$$SSR = SST - SSE = 68200 - 6394$$

$$r^2 = \frac{SSR}{SST} = \frac{68200 - 6394}{68200} \approx 0.906 = 90.6\%$$

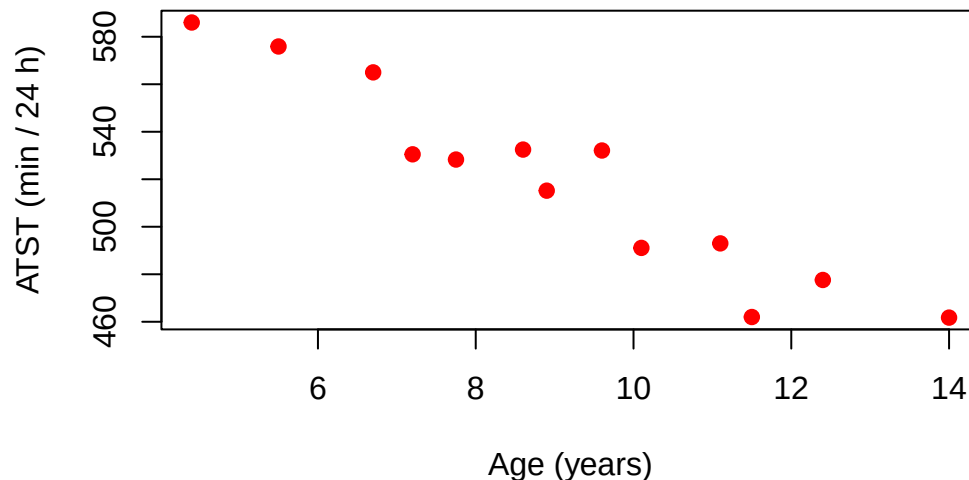
- c. About 90.6% of the variance in coffee sales is explained by the model.

Exercise 10 (11 points).

A group of 13 children and adolescents (considered healthy) participated in a psychological study designed to analyze the relationship between age and average total sleep time (ATST). To obtain a measure for ATST (in minutes), recordings were taken on each subject on three consecutive nights and then averaged. The results obtained are displayed in the following table.

Age (years)	ATST (min / 24 h)
4.40	586.00
14.00	461.75
10.10	491.10
6.70	565.00
11.50	462.00
9.60	532.10
12.40	477.60
8.90	515.20
11.10	493.00
7.75	528.30
5.50	575.90
8.60	532.50
7.20	530.50

The data is shown in the following plot. The r code output is also shown below.



```
fit <- lm(ATST ~ AGE)
summary(fit)
```

Call:

```
lm(formula = ATST ~ AGE)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-23.011	-9.365	2.372	6.770	20.411

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	646.483	12.918	50.05	2.49e-14 ***
AGE	-14.041	1.368	-10.26	5.70e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.15 on 11 degrees of freedom

Multiple R-squared: 0.9054, Adjusted R-squared: 0.8968

F-statistic: 105.3 on 1 and 11 DF, p-value: 5.7e-07

`anova_alt(fit)`

Analysis of Variance Table

	Df	SS	MS	F	P
Source	1	18220.5	18221	105.33	5.7e-07
Error	11	1902.8	173		
Total	12	20123.4			

- The psychologists were interested in the correlation between ATST (y) and AGE (x). Below is the scatterplot for these two variables. Do you observe a trend?
- The regression of ATST on Age R output is as follows. Calculate the coefficient of correlation r relating ATST y to AGE x from the R output above (round to 3 decimal places).
- Interpret the coefficient of correlation r you got in part (b). That is, is the linear relationship weak or strong? Is it a positive linear relationship or negative?
- Conduct a test to determine whether ATST y is correlated with AGE x . Use $\alpha = .01$. What are your null and alternative hypotheses?
 - $H_0 : \beta_1 \neq 0, H_a : \beta_1 = 0$
 - $H_0 : \beta_0 = 0, H_a : \beta_0 > 0$
 - $H_0 : \beta_1 = 0, H_a : \beta_1 \neq 0$
 - $H_0 : \beta_0 = 0, H_a : \beta_0 \neq 0$
 - $H_0 : \beta_1 > 0, H_a : \beta_1 = 0$
- Conduct a test to determine whether ATST y is correlated with AGE x . Use $\alpha = .01$. What is your conclusion?
 - As p -value is greater than $\alpha = .01$, we fail to reject the null hypothesis.
 - As p -value is smaller than $\alpha = .01$, we fail to reject the null hypothesis.
 - As p -value is greater than $\alpha = .01$, we reject the null hypothesis.
 - As p -value is smaller than $\alpha = .01$, we reject the null hypothesis.
- Using the conclusion drawn in the previous step, determine whether there is sufficient evidence to support the claim of a linear correlation between the ATST and the AGE. Choose the correct answer.
 - We conclude there is sufficient evidence to indicate the nonlinear correlation between the ATST and AGE at $\alpha = .01$.
 - We conclude there is not enough evidence to indicate the linear correlation between the ATST and AGE at $\alpha = .01$.
 - We conclude there is not enough evidence to indicate the nonlinear correlation between the ATST and AGE at $\alpha = .01$.
 - We conclude there is sufficient evidence to indicate the linear correlation between the ATST and AGE at $\alpha = .01$.
- Find $\hat{\sigma}^2$, the variance of the random error term in the model.

- h. $SSE = ?$ (round to 2 decimal places)
- i. $MSE = ?$ (round to 2 decimal places)
- j. $SE = ?$ (round to 2 decimal places)
- k. Give a practical interpretation of the value of s .

Solution:

- a. There is an approximately negative linear trend.
- b. $-\sqrt{0.9054} \approx -0.952$
- c. A strong negative linear relationship.
- d. C
- e. D
- f. D
- g. 173
- h. 1902.80
- i. 173
- j. $\sqrt{173} \approx 13.15$
- k. It is an estimate of the standard deviation of the random error term. Roughly, about 68% of the observed responses are expected to fall within s of the fitted value, and about 95% within $2s$ (assuming the error term is approximately normal).

Exercise 11 (4 points).

Scientists conducted a series of trials to investigate the feeding habits of a kind of birds. They were deprived of food until their guts were empty, then were allowed to feed for 6 hours. For each of 42 feeding trials, the change in the weight of the birds after 2.5 hours was recorded as a percentage of initial weight. The other variable recorded was digestion efficiency (measured as a percentage).

The data for selected feeding trials are listed in the table below.

Trial	Weight Change (%)	Digestion Efficiency (%)
1	-6.0	0.0
2	-5.0	2.5
3	-4.5	5.0
4	0.0	0.0
5	2.0	0.0
38	9.0	59.0
39	12.0	52.5
40	8.5	75.0
41	10.5	72.5
42	14.0	69.0

```
fit <- lm(weight_change ~ DIGEFF, data=dat)
summary(fit)
```

Call:

```
lm(formula = weight_change ~ DIGEFF, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.5501	-0.3057	0.6570	1.1310	3.7291

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.72912	0.55135	-3.136	0.00321 **
DIGEFF	0.21039	0.01331	15.802	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.956 on 40 degrees of freedom
Multiple R-squared: 0.8619, Adjusted R-squared: 0.8585
F-statistic: 249.7 on 1 and 40 DF, p-value: < 2.2e-16

```
anova_alt(fit)
```

Analysis of Variance Table

	Df	SS	MS	F	P
Source	1	955.34	955.34	249.7	8.5337e-19
Error	40	153.04	3.83		
Total	41	1108.38			

```
coef <- coefficients(summary(fit))  
coef_CI <- confint(fit, level=0.90)  
cbind(coef, coef_CI)
```

	Estimate	Std. Error	t value	Pr(> t)	5 %	95 %
(Intercept)	-1.7291172	0.55135210	-3.13614	3.205569e-03	-2.6575120	-0.8007224
DIGEFF	0.2103902	0.01331432	15.80180	8.533747e-19	0.1879709	0.2328095

```
new_digeff <- data.frame(DIGEFF = c(27.5, 35))  
CI <- predict(fit, new_digeff, interval='confidence', level=0.90)  
cbind(new_digeff, CI)
```

	DIGEFF	fit	lwr	upr
1	27.5	4.056613	3.523681	4.589545
2	35.0	5.634539	5.126261	6.142818

```
PI <- predict(fit, new_digeff, interval='prediction', level=0.90)  
cbind(new_digeff, PI)
```

	DIGEFF	fit	lwr	upr
1	27.5	4.056613	0.720138	7.393088
2	35.0	5.634539	2.301913	8.967165

Snow geese feeding trial R output.

- Find a 90% confidence interval for the true slope of the line.
- Based on the 90% confidence interval from the previous question, what can you conclude?
 - The confidence interval contains 0. Then, β_1 is significantly different from 0.
 - The confidence interval contains 0. Then, β_1 is not significantly different from 0.
 - The confidence interval doesn't contain 0. Then, β_1 is significantly different from 0.
 - The confidence interval doesn't contain 0. Then, β_1 is not significantly different from 0.
- The researcher wants to estimate the average weight change of all the birds with digestion efficiency 35% (i.e., DIGEPP = 35). From the R output, find the values of the interval at the 90% confidence level that is desired by the researcher.
- Give a practical interpretation of the interval in part (c).

Solution:

- (0.093, 0.190) (0.1879709, 0.2328095)
- C
- (0.714, 2.847) (5.1262612, 6.1428177)
- We are 90% confident that the true mean weight change for the birds with digestion efficiency 35% lies within this interval.